

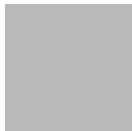
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Quest for the New Standard PSI IOC Platform

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- Accelerator facilities at PSI
- Motivation
- Goal
- Used platforms
- Requirements
- Hardware – possible ways to go
- Software
- Performance measurements
- Conclusions
- Acknowledgements

- **HIPA – High Intensity Proton Accelerator**
 - Over 40 years
 - Before: hp743, LynxOS, custom control system (home made)
 - Since 2001: MVME5100
 - Since 2011: VxWorks, EPICS
- **SLS – Swiss Light Source**
 - 20 years
 - VxWorks, MVME2304 + MVME5100, EPICS
- **PROSCAN – PROton SCANning Cancer Treatment Facility**
 - 10 years
 - Before: MVME5100, LynxOS, custom control system (home made)
 - Since 2008: VxWorks, EPICS
- **SwissFEL – Swiss Free Electron Laser**
 - Comming into operation just now
 - IFC1210 (IOxOS, VME), RTLinux, EPICS

- Equipment getting old
 - No replacements available (End of Life)
 - Performance can not satisfy new demands
 - Need higher CPU performance
 - System memory (RAM) speed should be higher
 - Higher bus speed needed (VME too slow)
 - New FPGA applications exceed available resources
 - Need bigger FPGA
 - Need more general purpose I/O
 - Need more optical links

- We really need new platform
- Only one if possible
 - Satisfy the needs of different groups
 - Can be used for a wide range of applications
- Should be scalable
 - From low demands, low cost (Digital PowerSupply, simple ADC/DAC apps, ...)
 - To high demands, higher cost (Low Level RF, DBPM, ...)
- Next Processing Platform (NPP) project was started
 - To gather requirements from different groups
 - Take decision

Used Platforms

- All four facilities are VME based
- IOCs
 - MVME5100/VxWorks with VME boards
 - No onboard FPGA and no FMC
 - With Simple VME I/O boards
 - And few VME boards with own FPGA
 - IFC1210/RTLinux with VME boards
 - Onboard Kintex Ultrascale FPGA, T2081 octacore CPU
 - With Simple VME I/O boards
 - And few VME boards with own FPGA
- All running EPICS as control system

Requirements

- Low latency loops (ADC -> DAC below $1\mu\text{s}$)
- 20 or more general purpose I/O
- FPGA at least 200K logic elements
- At least 8 Gigabit links
- Two Gigabit Ethernet links
- Long term availability
- VME form factor (nice to have option)
- Cost and effort saving
 - Can reuse existing VME transition boards (nice to have)
 - Can reuse existing FMC cards
 - Can reuse developed software and firmware
- All groups seem to favor Zynq UltraScale+

Ways to go

- Already available platforms
- Silicon on Module (SoM) Solution
- Silicon on Chip (SoC) Solution
- Complete commercial solution

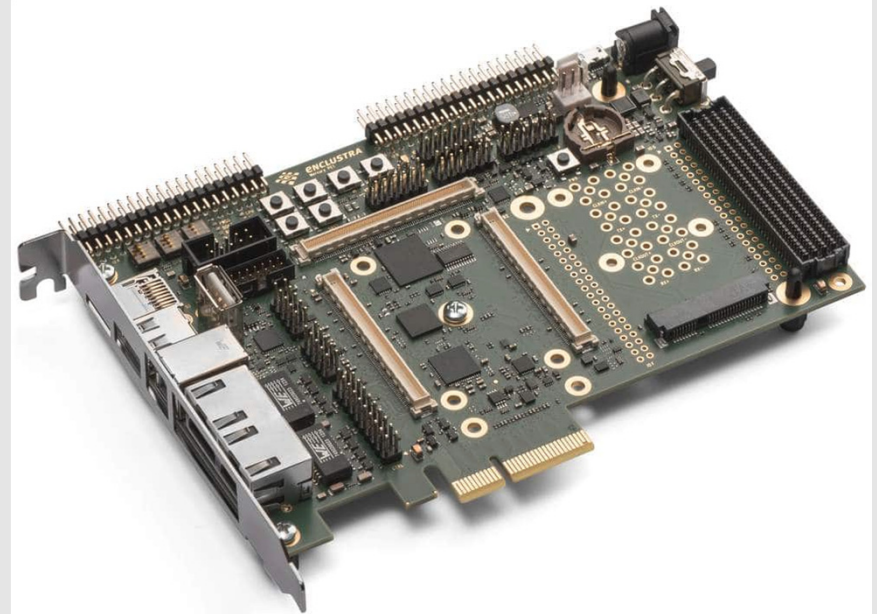
Ways to go – Use available platforms

- IFC1410/IFC1420 (IOxOS, MTCA, developed for ESS in Sweden)
- IFC1211 (IOxOS, VME)
- But, it is not Zynq UltraScale+



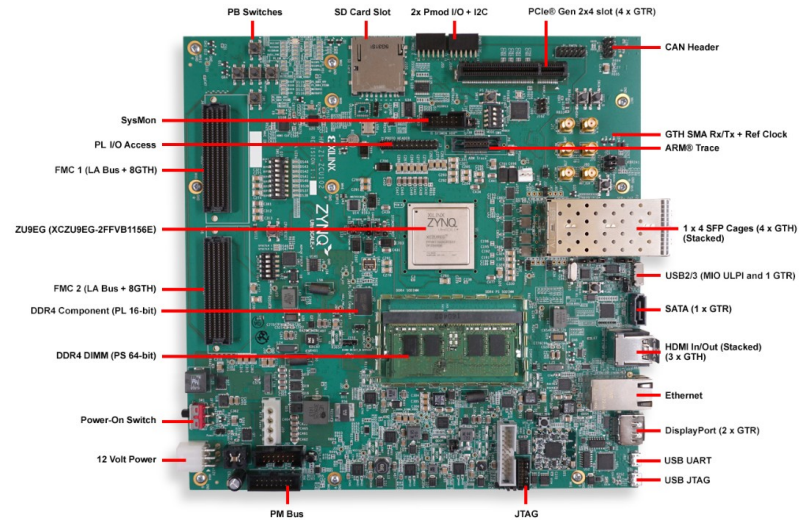
Ways to go – SoM solution

- Can buy, available in different speed/size grades (Enclustra)
- Needs carrier boards to be developed (ourselves or commercial partner)
- Can be made for VME or MTCA
- Or as bus-less system, own design
- It is Zynq UltraScale+



Ways to go – SoC solution

- Needs complete own design
 - Maybe based on already started PSI DBPM3
- Need more time, would probably cost more
- It is Zynq UltraScale+
- Used ZCU102 Reference board for test



Ways to go – Comercial MTCA solution

- Considering MTCA products (Vadatech)
 - just recently started investigating
- Zynq UltraScale+ based

Ways to go - Decision

- Decision is not made yet
- Will be made up to end of this year
- Will be Zynq UltraScale+ based

Software – IFC1410/IFC1420 and IFC1211

- Already all done
- Manufacturer provided u-boot, slightly modified
- Manufacturer provided Device Tree
- RTLinux Kernel
 - same version as for IFC1210
 - Needs „Tosca“ system firmware driver
 - we adapted it for this boards
- Root File System: Freescale NXP SDK 2.0 (Yocto based)
- EPICS 3.14.12, 3.15.5, 3.16.1 and 7.0.1 already available

- Already done (for available boards)
- U-boot and Device Tree
 - Manufacturer provided u-boot, slightly modified
 - Manufacturer provided Device Tree, slightly extended
- RTLinux Kernel
 - Planning to have one common kernel, manufacturer based
 - We have RT-patched it since manufacturers do not support RT
 - Works – but problems with Ethernet driver, ...
 - So still using manufacturers kernels
- Root File System
 - Using common Yocto Xilinx 2018.1 root file system
- EPICS 3.14.12, 3.15.5, 3.16.1 and 7.0.1 already available

Software – custom and commercial solutions

- To be done as soon as boards available
- Should be no big issue

Performance measurements

- Performance is not really important factor
- Approximate values – just to know what we can expect

Platform:	IFC1210	IFC1211	UltraScale+
Operation	(MB/s)	(MB/s)	(MB/s)
memcpy	200	500	2000

Platform:	IFC1210	IFC1211	UltraScale+
Test	(MFLOPS)	(MFLOPS)	(MFLOPS)
linpack	220	520	530

Platform:	IFC1210	IFC1211	UltraScale+
Operation	(μs/op)	(μs/op)	(μs/op)
sqrt	0.335	0.215	0.050
pow	0.780	0.655	0.360
exp	0.345	0.325	0.150
log	0.765	0.325	0.150
atan/asinh/acosh	0.350	0.290	0.160
sin/cos	0.280	0.290	0.135

Conclusions

- The new processing platform will be Zynq UltraScale+ based
 - We have to decide which one very soon
- Software is not show stopper

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**Thank you for your
attention**

Q & A

